

PRODUCT REQUIREMENTS DOCUMENT

PRISM

AI-Powered Automotive Procurement & Supplier Risk Management System

Academic Year: 2026–2027

1. Product Overview

PRISM is an AI-powered procurement and supplier risk intelligence system designed for the automotive industry.

It connects:

- Production requirements
- BOM
- Inventory
- Supplier profiles
- Quotations
- Historical prices
- Supplier performance
- Delivery information

PRISM helps procurement teams determine:

1. What needs to be procured?
2. Who can provide it?
3. What could go wrong?
4. Why is the supplier becoming risky?
5. What should we do?
6. How can we negotiate better?

PRISM uses a **Master Procurement Agent** to coordinate five specialized workflows:

- Planning & Inventory
- Supplier Intelligence
- Risk & Monitoring

- Decision & Prescriptive
- Negotiation

2. Problem Statement

Automotive procurement information is often distributed across spreadsheets, ERP systems, emails and other sources. Manual quotation comparison and historical analysis can be time-consuming, while emerging supplier risks may only become visible after affecting delivery or production.

For example:

Sept 10 → Sept 13 → Sept 17 → Sept 20

Repeated delivery postponements may indicate deteriorating supplier performance and potential delivery risk.

Therefore, PRISM aims to combine multiple signals, identify emerging risks early, explain the evidence, estimate impact and support preventive action.

3. Product Vision

To transform supplier and procurement data into **early risk insights, evidence-based recommendations and better purchasing decisions**.

Reactive Procurement → Proactive Procurement Intelligence

4. Goals

- Identify emerging supplier risks early.
- Improve supplier evaluation and selection.
- Reduce manual quotation comparison.
- Provide historical price intelligence.
- Identify savings and negotiation opportunities.
- Continuously monitor supplier conditions.
- Provide explainable AI recommendations.
- Maintain human control over important decisions.

5. Key Stakeholders

Stakeholder	Main Responsibility
Procurement Manager	Procurement decisions, supplier evaluation, negotiation
Procurement Buyer/Analyst	RFQs, quotations and analysis

Stakeholder	Main Responsibility
Supplier	Quotations, negotiation responses, delivery updates
Operations	Supplier verification and qualification
Finance/Approver	Cost and commercial approval
Production/Supply Chain	Production, component and inventory requirements

6. Pain Points

Problem	PRISM Response
Scattered supplier information	Centralized procurement intelligence
Manual quotation comparison	Automated structured comparison
Difficult historical price analysis	Historical price intelligence
Late supplier-risk identification	Risk prediction and monitoring
Multiple signals difficult to combine	Multi-factor analysis
Risk causes unclear	Evidence-based risk explanation
Cheapest supplier may not be suitable	Multi-factor supplier evaluation
Manual supplier follow-up	Continuous monitoring
Negotiation depends on experience	AI negotiation assistance
New suppliers lack history	Cold-start handling

Product Principle

PRISM does not simply identify the cheapest supplier.

It considers **price + delivery + quality + capacity + lead time + performance + risk + commercial conditions**.

7. Functional Requirements

FR1 — Authentication & RBAC

Secure login and role-based access control.

FR2 — Procurement Dashboard

Display:

- Procurement requirements
- Supplier information
- Quotation comparisons
- Risk alerts
- Supplier scores
- Price variance
- Delivery performance
- Potential savings
- AI recommendations

FR3 — Planning & Inventory

Analyze:

- Customer orders
- Production requirements
- BOM
- Current inventory
- Incoming supply
- Required dates

to determine procurement requirements.

FR4 — Supplier Intelligence

Evaluate supplier suitability using price, delivery, quality, capacity, lead time and historical performance.

FR5 — Risk Prediction

Assess delivery, inventory, capacity, quality, supplier-performance and commercial risks.

FR6 — Decision Support

Generate and evaluate candidate procurement actions.

FR7 — Negotiation Assistance

Provide target prices, counteroffers, strategies and message assistance.

FR8 — Monitoring

Monitor supplier, inventory, PO, shipment and delivery information and trigger re-analysis after significant changes.

8. Non-Functional Requirements

Attribute	Requirement
Performance	Timely dashboard, quotation and risk analysis
Scalability	Support increasing suppliers and transactions
Security	Authentication, RBAC and protected data
Reliability	Consistent procurement information
Explainability	Evidence, reasoning, confidence and impact
Auditability	Record important actions and decisions
Maintainability	Modular architecture and workflows
Interoperability	API-based communication
Usability	Understandable insights for non-technical users

9. Design Constraints

- SNS Agent Workbench is used for agent orchestration.
- FastAPI acts as the central API gateway.
- PostgreSQL acts as the source of truth.
- React does not directly communicate with AI agents.
- Important purchasing decisions require human review.
- Missing information must not be fabricated.

10. Supplier Management

PRISM maintains supplier profiles containing:

- Supplier ID
- Supplier name
- Contact information
- Production capacity
- Supplier status
- Qualification information

- Delivery history
- Quality history
- Supplier performance
- Operational information

This information supports Supplier Intelligence and Risk & Monitoring.

11. RFQ Management

Users can create and manage Requests for Quotations.

RFQ Information

- Component
- Quantity
- Required delivery date
- Specifications
- Selected suppliers
- RFQ status

RFQ Lifecycle

Created → Sent → Supplier Response → Quotation Received → Under Evaluation

12. Quotation Processing

Quotations may be received through:

- Excel/CSV
- API
- PDF documents

Minimum Quotation Information

- Supplier
- Component/Product
- Quantity
- Unit Price
- Currency
- MOQ

- Lead Time
- Promised Delivery Date
- Delivery Terms
- Payment Terms
- Warranty
- Commercial Conditions
- Quote Date
- Quote Version

Processing Flow

Upload → Document Processing/OCR → AI Extraction → Structured Data → Validation → Database

Important Rule

If information is missing or ambiguous, PRISM must **flag it for validation rather than inventing a value.**

13. Supplier Intelligence

PRISM evaluates suppliers using:

- Price
- Delivery performance
- Quality
- Capacity
- Lead time
- Historical performance
- Supplier status
- Commercial terms

The output includes supplier scores and comparison information.

14. Quotation Comparison

The comparison dashboard evaluates:

Parameter

Price

Parameter

Quantity

Delivery Time

Payment Terms

Warranty

Commercial Conditions

Supplier Performance

Supplier Score

Example

Supplier A: ₹125, 5,000 units, 20 days

Supplier B: ₹130, 5,000 units, 15 days

Supplier C: ₹122, 4,000 units, 30 days

The comparison supports the procurement manager rather than automatically making the final decision.

15. Historical Price Intelligence

PRISM compares current quotations with:

- Previous purchase price
- Current quotation price
- Previous quantity
- Current quantity
- Supplier
- Purchase date
- Price trends

Example

Previous ECU price: **₹120/unit**

Current quotation: **₹135/unit**

Difference: **₹15/unit**

This may indicate a negotiation opportunity after considering quantity, supplier context and market conditions.

16. Procurement KPIs

- Supplier Score
- Price Variance
- Historical Price Difference
- Potential Savings
- Delivery Performance
- Lead-Time Performance
- Inventory Coverage
- Risk Score
- Negotiation Opportunity
- Procurement Decision Time
- Quotation Processing Time

17. Risk & Monitoring

Risk Categories

1. Delivery Risk
2. Material/Stock Shortage Risk
3. Capacity Risk
4. Quality Risk
5. Supplier Performance Risk
6. Commercial/Price Risk

Risk Signals

- Frequent delivery postponements
- Lead-time variability
- Quality rejection
- Quantity shortages
- Capacity constraints
- Supplier status changes

- Low inventory coverage
- Incoming supply delays
- Historical price changes
- Production pressure
- Missing supplier updates

Important Principle

A single signal is **not automatically proof of supplier failure**.

PRISM evaluates multiple signals together with their context.

18. Risk Prediction

Each risk assessment contains:

- Risk Type
- Risk Score
- Severity
- Evidence
- Confidence
- Potential Impact

Example

Supplier: ABC Components

Risk: Delivery Risk

Severity: High

Evidence: Multiple delivery postponements

Impact: Component shortage before production date

Confidence: Medium

The system explains **why** a supplier is becoming risky instead of showing only a numerical score.

19. Decision & Prescriptive Workflow

The workflow combines:

- Planning & Inventory
- Supplier Intelligence
- Risk & Monitoring
- Historical data

- Procurement constraints

Possible actions:

- Continue with supplier
- Negotiate earlier delivery
- Split order
- Select alternative supplier
- Increase safety stock

Actions are evaluated using cost, delivery, quality, risk, inventory and production impact.

20. Negotiation Workflow

PRISM supports procurement managers with:

- Quotation analysis
- Historical price comparison
- Target price suggestions
- Counteroffer suggestions
- Negotiation strategy
- Message drafting
- Supplier response analysis

Example

Current quotation: **₹135/unit**

Historical price: **₹120/unit**

Possible target: **₹125/unit**

Possible counteroffer range: **₹122–₹125/unit**

These are **decision-support suggestions**. The procurement manager reviews and sends the negotiation communication.

21. Master Procurement Workflow

The Master Procurement Agent coordinates the five specialized workflows.

Specialist Workflows

1. Planning & Inventory

2. Supplier Intelligence
3. Risk & Monitoring
4. Decision & Prescriptive
5. Negotiation

Master Flow

User Request



Master Procurement Agent



Relevant Specialized Workflows



Aggregated Intelligence



Recommendation



Human Approval

The Master Agent routes information, aggregates results and triggers re-analysis when required.

22. New Supplier / Cold Start

New suppliers may have no historical performance.

PRISM does not fabricate historical data.

Instead, it evaluates available:

- Current quotation
- Offered quantity
- Promised delivery
- Lead time
- Capacity
- Qualification
- Market conditions

- Commercial terms

The system displays:

“Insufficient Historical Evidence”

or provides a lower-confidence assessment.

As transactions occur, actual supplier performance is collected.

23. Human-in-the-Loop

AI can:

- Analyze
- Predict
- Explain
- Recommend
- Suggest negotiation strategies
- Draft messages

Humans:

- Review recommendations
- Approve important decisions
- Modify recommendations
- Reject unsuitable recommendations
- Approve commercial decisions

Finance approval is required where applicable before PO generation.

24. Continuous Monitoring

PRISM monitors:

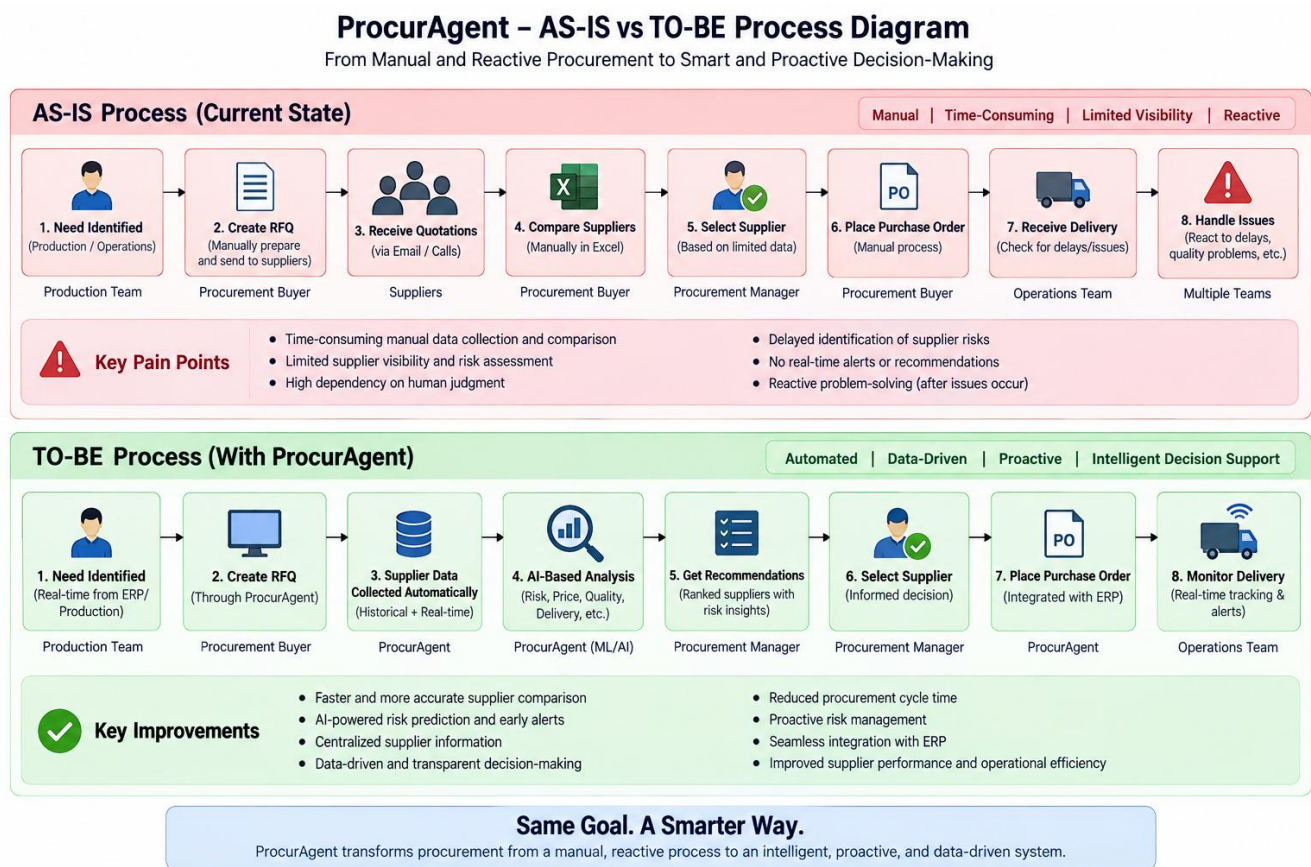
- Supplier performance
- Inventory
- Purchase orders
- Shipment milestones
- Delivery milestones
- Supplier updates

- Reported issues

Re-analysis

New Event → Risk Analysis → Decision Analysis → New Recommendation

25. AS-IS Process & TO – BE

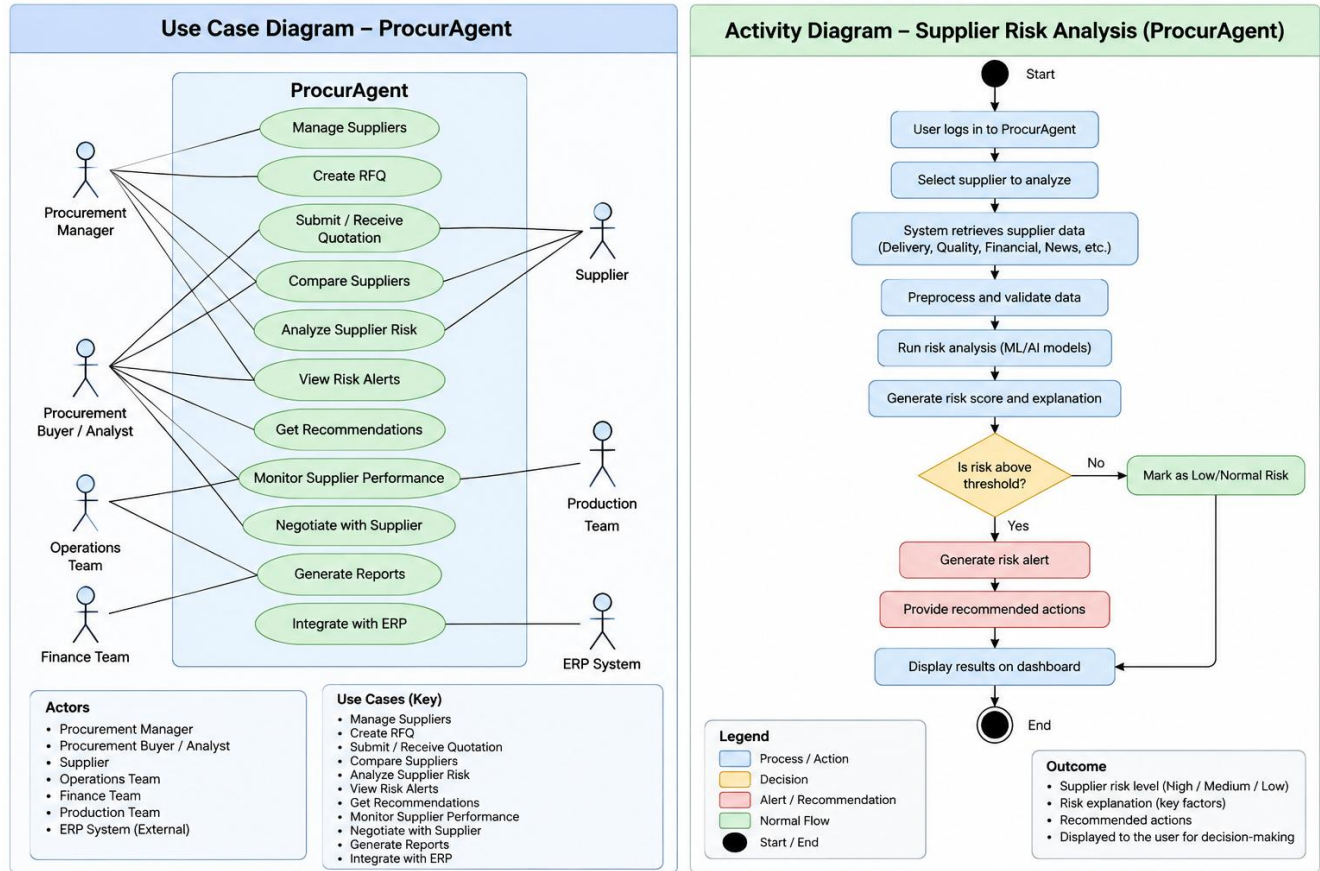


26. User Journey

Stage	User Action	System Output
Dashboard	Review KPIs	Procurement overview
Need Identification	Review requirements	Procurement requirement
RFQ	Create RFQ	RFQ sent
Quotation	Upload quotation	Structured quotation
Supplier Analysis	Compare suppliers	Supplier ranking
Risk Analysis	Review risks	Risk assessment

Stage	User Action	System Output
Decision	Review alternatives	Recommendation
Negotiation	Review target/counteroffer	Negotiation support
Approval	Approve/modify/reject	Authorized decision
PO	Review PO	Purchase order
Monitoring	Track supplier	Alerts
Re-analysis	Respond to events	Updated recommendation

27. Key Use Cases



28. System Architecture

Users / External Systems

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React + Vite Frontend

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FastAPI API Gateway

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SNS Agent Workbench

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Python ML / Intelligence

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PostgreSQL Database

Responsibilities

React/Vite: UI, dashboards, RFQs, comparison, risk and approvals.

FastAPI: Authentication, RBAC, APIs, database communication and integrations.

SNS Workbench: Agent orchestration, routing, reasoning and human-in-the-loop.

Python: ML models, forecasting and analytical processing.

PostgreSQL: System of record.

29. AI & ML

ML

Used for potential:

- Delivery-delay prediction
- Material-shortage prediction
- Quality-risk prediction
- Supplier-performance deterioration
- Capacity-risk assessment
- Price trend/anomaly analysis

AI Agents

Used for:

- Workflow coordination
- Context understanding
- Reasoning
- Risk explanation

- Recommendations
- Negotiation assistance
- Event-driven re-analysis

ML predicts what may happen; AI agents reason about what should happen next.

30. Data Requirements

PRISM uses:

- Customer orders
- Production plans
- BOM
- Component requirements
- Inventory
- Incoming supply
- Supplier profiles
- Supplier performance
- Delivery/quality history
- Supplier capacity
- Quotations
- Historical prices
- Market/reference information
- Shipment/delivery milestones
- Risk assessments
- AI recommendations
- Approval and audit records

Prototype data may use realistic synthetic automotive procurement data because real procurement information is generally confidential.

31. Major Database Entities

- User
- Supplier

- Product/Component
- RFQ
- Quotation
- Purchase Order
- Inventory
- Inventory Transaction
- Supplier Performance
- Historical Purchase
- Risk Assessment
- Negotiation
- Recommendation
- Approval
- Shipment/Delivery

32. Core Business Rules

1. Competitor supplier information must not be exposed.
2. Missing quotation information must not be fabricated.
3. New supplier history must not be fabricated.
4. AI recommendations require supporting evidence.
5. AI must communicate uncertainty.
6. Important commercial decisions require human review.
7. Finance approval is required where applicable.
8. Supplier acknowledgement is required before PO activation.
9. Important decisions must be auditable.
10. PostgreSQL is the source of truth.
11. FastAPI is the central API gateway.
12. React does not directly communicate with AI agents.
13. Specialized workflows remain within their responsibilities.
14. Master Agent coordinates the workflows.

33. MVP

Must Have

Authentication/RBAC, Supplier Management, Product Management, RFQ, Quotation Ingestion, AI Extraction, Validation, Comparison, Historical Price Comparison, Supplier Scoring, Dashboard, Basic Risk Analysis, AI Recommendations, Negotiation Assistance and Master/Specialized Workflows.

Should Have

Real-time monitoring, milestone tracking, advanced risk prediction, alerts and advanced price intelligence.

Could Have

Market intelligence, scenario simulation, advanced optimization and additional integrations.

Out of Scope

Fully autonomous purchasing, automatic supplier termination, direct financial transactions and physical factory control.

34. Success Criteria

PRISM should demonstrate that:

1. Procurement requirements are correctly identified.
2. Supplier quotations can be structured.
3. Multiple quotations can be compared.
4. Historical prices support benchmarking.
5. Supplier performance can be evaluated.
6. Emerging supplier risks can be identified.
7. Risk explanations contain evidence.
8. AI generates actionable recommendations.
9. Negotiation suggestions can be generated.
10. Humans can approve or reject recommendations.
11. New suppliers are handled without fabricated history.
12. Significant events can trigger re-analysis.

35. Differentiation

Traditional procurement systems primarily focus on recording and managing transactions.

PRISM adds an intelligence layer:

Procurement Data + Supplier Data + Historical Data + Inventory Data



AI / ML Analysis



Risk Detection



Impact Analysis



Recommended Action



Human Decision

PRISM therefore acts as an **intelligent procurement co-pilot**.

Final Outcome

Earlier risk visibility + faster procurement analysis + better supplier evaluation + stronger negotiation preparation + explainable AI-assisted decisions.

